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CASE 1: Scanning a standard colored plastic sample

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[STAGE 1] Scanning sample with Near-Infrared (NIR) sensor...

[NIR RESULT] Confidence: 89.87%

--- FINAL CLASSIFICATION REPORT ---

Sensor Used: NIR
Predicted Polymer: PVC
Prediction Confidence: 89.87%
True Class: PVC
Fallback Triggered: False
Status: SUCCESS [OK]

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CASE 2: Scanning a black plastic sample (e.g., carbon black colored)

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[STAGE 1] Scanning sample with Near-Infrared (NIR) sensor...

[NIR RESULT] Signal strongly absorbed. Likely black plastic.

[NIR RESULT] Confidence: 15.00%

[NIR WARNING] Confidence (15.00%) below threshold (85%).

[STAGE 2] NIR scan failed or inconclusive. Activating MIR backup sensor...

--- FINAL CLASSIFICATION REPORT ---

Sensor Used: MIR (Backup)
Predicted Polymer: HDPE
Prediction Confidence: 99.00%
True Class: HDPE
Fallback Triggered: True
Status: SUCCESS [OK]